

The EdTech platform, JoveNotes, is open source and hosted on GitHub. I have provided a summary of all the source repositories at the end of this article.

A high spirited but short lived start

Fueled by the desire to pass on my life's learning and dreams of building a great father-son relationship, I, like every parent, dived into the role. This was back in 2008.

Sadly, the reality couldn't have been more contrasting. Business travels, office work, and other domestic priorities soon ate into the resolve. Toil and turmoil of the day manifested as fragile egos and volatile emotions on the study table. It seemed all our time was getting consumed in playing catch-up, leaving little time to explore the joys of knowledge. My role soon devolved from a mentor to a coach, then a trainer, and finally a monitor. The fun was gone, and it was not funny anymore.

Denial and attribution kicked in soon afterwards. I joined the herd of pained parents in attributing our anguish to the flawed education system.

Arresting the vicious downward spiral

It was high time to revisit the ambition. The apparent solution was to arrange for additional tuitions. But if the 'education system' is rotten, which I had started to believe, how would tuitions help? What is this 'education system', by the way? Is our education system built on a documented framework?

The information was not very forthcoming but not very difficult to find either. Finally, I came across the NCERT's national study on the school curriculum and the CISCE framework for the primary and upper primary curriculum.

A welcome change in perspective

It was an eye-opening experience. I was awed by the depth of thought process underpinning the architecture of our education system. Yes, I found some things, not to my liking, but I also started appreciating why things needed to be the way they are. Following were my five key takeaways.

- **Learning the subjects is not the end objective. They are, but a means to an end.** The end goal of our education system is to instil the required life skills of perseverance, accuracy, continuous learning, managing impulsivity, thinking flexibly, taking responsible risks, etc.
- **We use two major capability measurement yardsticks – retention & recollection, and problem-solving.** All subjects use these, though in different proportions. Math, for example, has a high ratio of applicability compared to, say, Social Studies, which requires a high degree of retention and recollection.
- **Humanities dominate the curriculum in the early years. The proportion of science and math slowly increases as we progress towards the upper primary.** This results in higher expectations of retention and recollection compared to problem-solving in early schooling years. There is a sharp jump in problem-solving expectations post 10th, especially for the science stream, which puts students at a disadvantage.
- **Although the design is explicit on the need to impart life skills, operationalization has lost the intent.** Our incessant desire to measure intelligence by exam scores has been our nemesis.
- **The volume of the curriculum for each class ‘looks’ excessively high; it’s not so.** For example, in class 7, there are 135 chapters spread across ten subjects, scheduled to be covered in a 36-week academic year. This volume looks vast because we are used to looking at curriculum class by class, sequentially. Instead, the curriculum builds bodies of knowledge in a gentle ramp across multiple years. For example, Algebra starts in class 6 and continues till 10th, each year adding only a few new concepts and spending a significant time rehashing contents taught earlier.

The new north star

I am glad I chose to dive deeper into the design of our education system. It gave me perspective and showed me areas where I could add value. There were four such areas.

- **Develop problem-solving skills without compromising on the wealth of humanities.** It was clear that my elder one would pursue sciences. Now that I knew of the capability chasm waiting for him post 10th, I could mentor him in advance on foundational problem-solving skills, like, accuracy, persistence, lateral thinking, metacognition, etc.
- **Focus on life skills not taught effectively at school.** The school is a great place to learn capabilities like teamwork, rigour and discipline, etc. However, life skills like imagining, innovating, thinking flexibly, managing impulsivity, and persisting are not taught effectively at school.
- **Synergize bodies of knowledge and maintain developmental context.** From the eyes of a child, the education system must look like a ninja obstacle course. We discretize education to achieve operational efficiencies. Teachers change with subjects, and every class has a new set of teachers. This discretization has resulted in a state-less education system, missing out on vital aspects of our education system – driving synergy between subjects and maintaining a child's developmental context from year to year.
- **Need of a hyper-efficient system.** All the above are good in theory unless we find a way to carve us more time. There are fortunately not many options – either we optimize on the effort by dilution or redirection (like we are used to), stress out the poor kid or leverage technology. I didn't want to do the former; my better half would raise hell if I hinted the second. And, hence, I followed the famous quote by Andy Weir – *"So, in the face of overwhelming odds, I'm left with only one option: I'm going to have to science the shit out of this."*

Why build when so many exist?

Before you ask – Yes, I did spend considerable time evaluating EdTech platforms and products. But, unfortunately, none worked for me because of three reasons.

First was the lack of contextualization. All the EdTech platforms I evaluated were content-rich but context-unaware. Schools teach at their own pace and from textbooks of specific publishers. For the platform to be of value, it needs to be in lockstep with the school, which hardly any platform did.

Second, I wanted the system to play a critical role by maintaining a multi-year context and adapting to the child's learning style. We would, hence, become more and more dependent on the platform as time went by. Guaranteed availability, lack of portability, and anxiety of platform lock-in in became an overwhelming concern.

Third, I wanted a rich stream of insights to help coach and mentor better. Most commercial platforms focused on colourful dashboards, which were of little value. I also wanted access to raw data to derive my insights, which was impossible with commercial platforms.

So, with unbridled enthusiasm, I told my wife – "*We need to build an automated coaching system.*" "*You think you can solve everything by writing code!*" boomed back along with a few other not-so-subtle feedback. So, for the next few days, risking marital harmony, I stealthily coded. The risk of disrupting marital peace paid off nicely as my wife, in due course, became a key contributor to the success of the platform.

Over the next twelve years, the system evolved from a mere dashboard to a full-fledged intelligent coaching system. It saw my elder one through his entire school and college education and has already seen my younger one through seven academic seasons. It gifted us hundreds of hours of quality time, helping us realize our dream of exploring the beautiful world of knowledge.

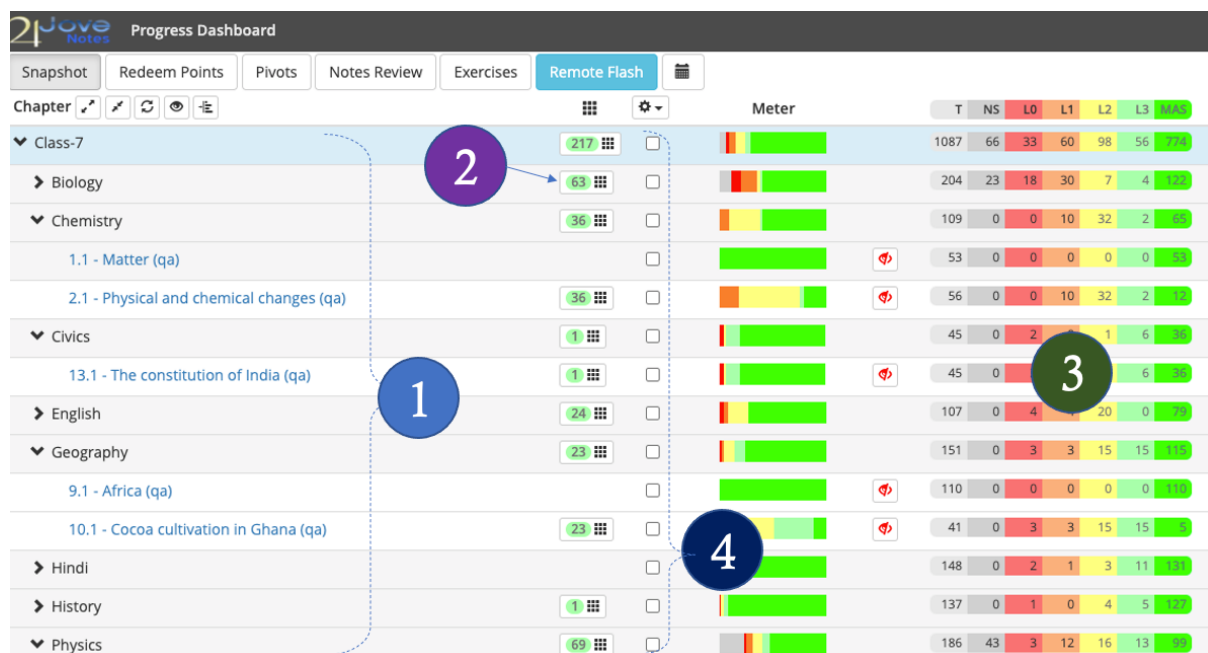
JoveNotes - The System

JoveNotes is architected to operate out of a Raspberry Pi comfortably. Broadly, the platform has five architectural building blocks.

- **The content manager** includes a custom-built high-speed scanner, a Git-based content management system and a custom-built IDE to edit digitized content.
- **JoveNotes compiler.** The platform uses a custom DSL (Domain Specific Language) to structure its input content. The compiler uses this DSL to compile the source into a canonical knowledge model on which it operates.
- JoveNotes web application presents the web-based responsive user interface.
- JoveNotes batch is a scheduled processor which does housekeeping work, including updates to its OLAP cube.
- Tools and utilities contain various utilities to automate many platform activities like creating content structure, automating the creation of digitized sources, etc.

JoveNotes has five key feature modules. Dashboard, Challenge-Response, Problem Solving, Reward Redemption and Reports.

Dashboard



The dashboard provides a single pane progress summary for the entire syllabus.

On the left (Section 1) is the collapsible tree of the chapters. JoveNotes can present the digitized metadata of each chapter either as a questionnaire or as an eBook. These digital notes are adaptive. The system dynamically adjusts the content of these notes based on the child's learning progress, ensuring that attention is directed only towards the body of knowledge needing attention. The content can also be filtered and printed out as exercise booklets for offline practice.

The screenshot displays the JoveNotes interface. On the left, a collapsible tree of chapters is visible. The central content area features a table comparing Parenchyma, Collenchyma, and Sclerenchyma tissues, followed by a list of challenges for identifying supporting tissue types. The right sidebar shows a lesson text titled '2. Swami and the Sum' with an illustration of a boy reading a book.

Category	Parenchyma	Collenchyma	Sclerenchyma
Shape	Spherical, thin-walled cells	Elongated cells with thickened cell walls	Thickened cell walls
Intercellular space	Large intercellular spaces	Little intercellular space	Little intercellular space
Maturity	Consists of living cells at maturity.	Consists of living cells at maturity.	Consists of dead cells at maturity.
Location	Found throughout the plant, in every soft part	Found in petioles, leaves and young stems	Found in old stems and roots
Function	Photosynthesis, respiration, storage, food, buoyancy in aquatic plants	Support, storage, photosynthesis	Support, storage, photosynthesis

Identify the type of supporting tissue:

- Fill in the blanks:
 - _____ is the second largest and the second most populated continent after _____.
 - The _____ divides Africa into two parts.
 - Mount _____, also known as Mt. _____, is the highest peak of Africa.
 - The South-Eastern plateau extends from _____ to the south of Africa.
 - The Nile river originates from _____, the largest lake in Africa.
 - The two main tributaries of the Nile meet at _____ in _____.
 - The _____ Dam on the Nile is used for irrigation, flood control and power generation.
 - The _____ river is also known as the Congo river.
 - The main tributaries of the Zaire river are the _____ and the _____.
 - Lake _____ is the third largest lake in the world after _____ and _____.
 - Lake _____ is a rift valley lake. It is the world's second deepest lake and the second largest lake of Africa.
 - _____ river of Africa is the longest river in the world.
 - The famous Victoria Falls lie on the _____ river.
 - The Temperate Grasslands of Africa are called the _____.
 - _____ is the lowest point in Africa.

JoveNotes generates challenges too from the same metadata. Using a spaced repetition (SR) algorithm, the system determines if a challenge is worth revisiting. Section 2 shows the qualifying challenge count for each topic. [Spaced repetition] is a well-researched, evidence-based learning technique to enhance retention and recollection abilities. Clicking on each button will load the exercise module for practising the challenges. Spaced repetition cuts down on the study effort by practising only those challenges that need attention.

Spaced repetition decides the practice intervals for each challenge based on the learning efficiency. The more the number of consecutive correct responses, the longer the gap till the next attempt. Challenges thus get classified based on their learning efficiencies - Not Started (NS), Level 0, 1, 2, 3, and mastered. Section 3 on the right shows the maturity concentration of challenges in each topic. On the left is a visual representation using stacked bar charts. This progress representation helps us let go of traditional timetable based fixed study schedules and enables us to prioritize our study schedules dynamically.

Users can launch a study session including exercises from multiple topics by selecting topics in section 4.

Challenge – Response

"Let's practice."

"Nooo, I don't want to study today."

"Let's practice only 15 questions?"

"Nooo"

"Let's put a timer. Fifteen questions or 5 minutes, whichever is faster."

"Ok.. but I don't want to do question-answers."

"Sure.. how about true-false and multiple-choice questions?"

"Yes, ok. We can also do some fill in the blanks if you like."

"Done! Let's start when you are ready."

Variations of the above exchange have become a common occurrence for me. Kids are kids because they are kids. Kids study best when they commit, and getting them to be part of decision making is the best way to secure their commitment.

JoveNotes presents an elaborate negotiation screen to configure the study sessions. Here we can configure the session by choosing upper bounds on the number of challenges, study time, and new challenges. We can also filter questions of a particular type, maturity level, difficulty level and current learning efficiency.



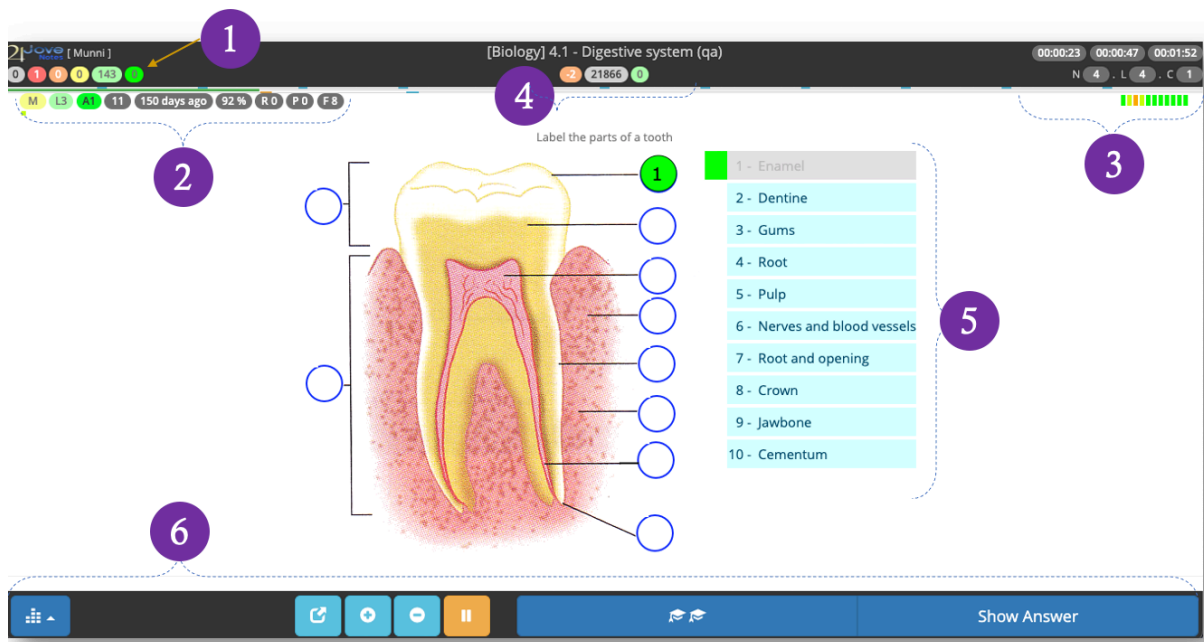
There are also various strategies for having the system auto-assemble challenges for practice. Spaced repetition algorithm is the default. Other choices include ascending or descending order of the learning efficiencies, descending order of recency of training, etc.

Some additional study strategies are 'Preparedness' and 'Retention', which involve machine learning. JoveNotes learns the retention efficiency of a student heuristically. It can, hence, predict a probability of answering the question correctly at any given point in time. Retention strategy can, thus, pick up questions based on ascending order of their current retention values. Similarly, given a future date (exam date), JoveNotes can predict a forward-looking probability of answering a challenge correctly on the exam day. Thus, the preparedness strategy can prioritize challenges in ascending order of their forward-looking preparedness probabilities.

Here are two compelling use cases. First, on exam eve, we can switch on the 'Efficiency (Hard)' strategy and scan through only those challenges which are below 90% efficiency level, which is hardly 5-6% of the study volume. No mad scrambling on exam eve! Second, JoveNotes can prioritize the questions according to the forecasted preparedness score and the exam schedule. Thus, providing for a cruise-controlled, hands-free preparation schedule.

'Engage fatigue buster' is another cool option. Once the session starts, JoveNotes tries to compute a 'fatigue' measure and presents easy or hard questions depending upon the fatigue level. Fatigue level is calculated based on multiple parameters, like efficiency level, difficulty level, the efficiency of answering last 'n' questions, responding time divergence from the predicted time, session duration, etc.

The configuration screen also displays the forecasted time to complete this session based on the choices made. Finally, the start button takes us to the practice screen, as shown below.



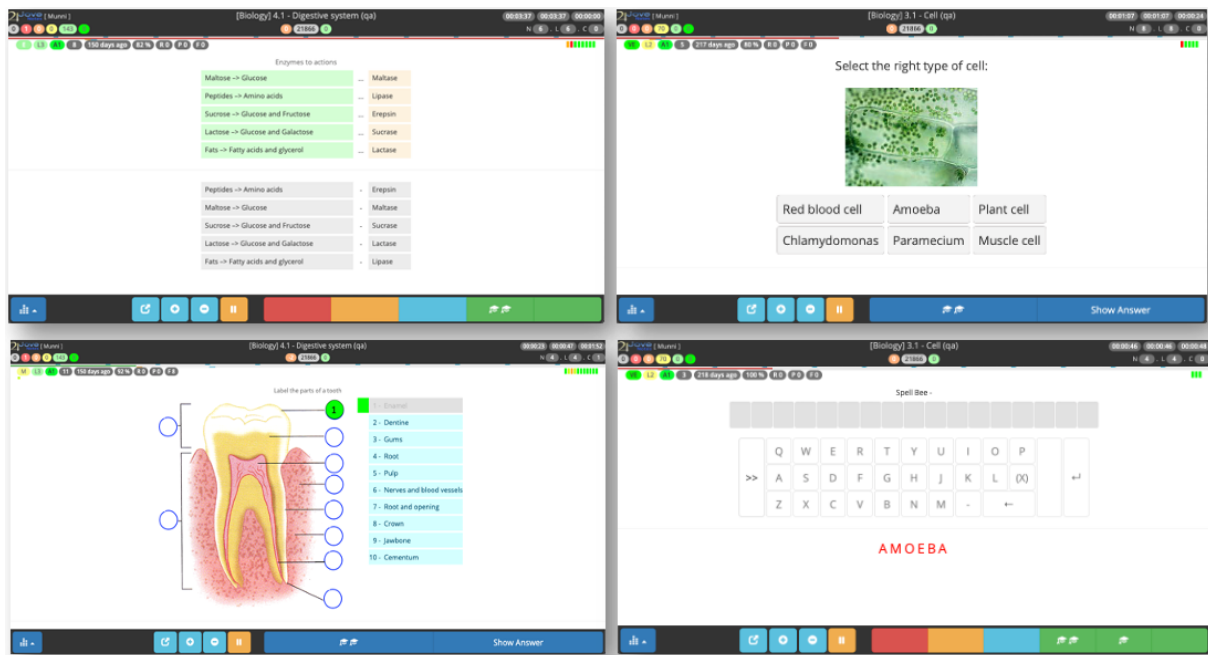
This is the challenge-response screen. It has six key sections. On the top left (section 1) is a view into the maturity concentration of challenges in the current topic.

Section 2 displays information like the difficulty level of this challenge, the maturity level, learning efficiency grade, number of attempts, number of days since the previous attempt, absolute learning efficiency, retention score, preparedness score and the current fatigue level.

Section 3 displays multiple clocks and timers – average time per challenge, total time spent in this session, and approximate time to completion. The number of challenges in this session is also displayed. Finally, the caterpillar bar provides a visual representation of the performance for all attempts of this question — an invaluable insight for a mentor.

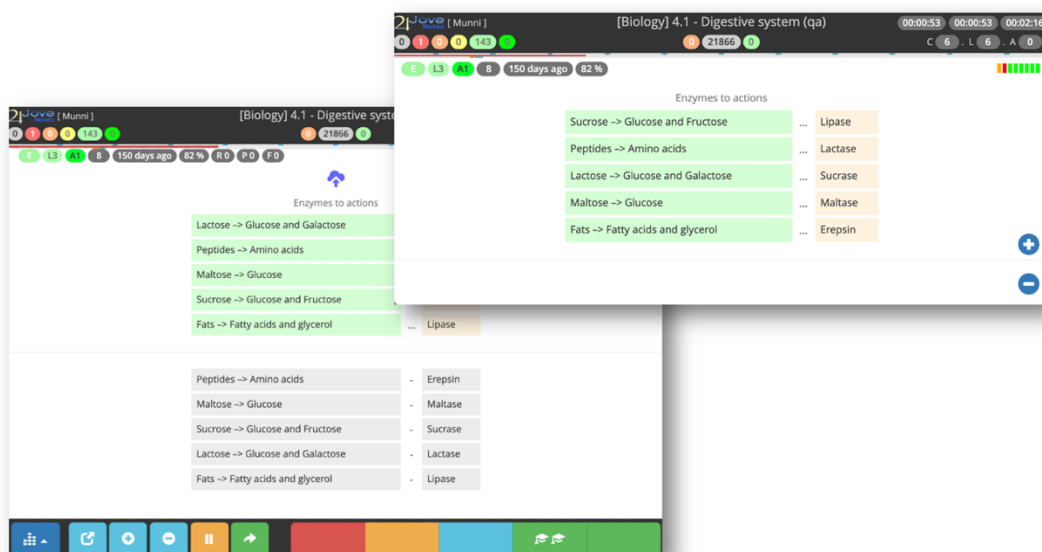
Section 4 hosts the reward counters. JoveNotes gamifies each question and awards reward/penalty points based on a logic catering for the state of maturity, the ratio of the response taken versus the projected optimal answer time, degree of correctness of the response and the difficulty level of the challenge. In addition, there is a separate module for reward point management.

Section 5 is the challenge presentation area where an interactable, audio-visual user interface presents each question based on its type. There are many challenge types – question-answer, matching, multi-choice, spell bee, image map, true-false, etc. The image below shows a sample of different question types.



Section 6 is where the dynamic controls are. There are controls for rating the answer, marking the question for review, adjusting zoom levels and many more. It is through these interactions that JoveNotes gets its learning inputs and adapts itself.

JoveNotes supports assisted learning which is more like a co-browsing experience where the coach and student share the same study session. The picture below shows the remote study setup. On the left is the coaching screen, while the screen on the right is the student screen. The coach navigates the study session while the student receives the flashed challenges. I have used this feature extensively to coach during my frequent travels.



Problem Solving

There is a reason why higher-order problem solving is introduced so late in the education curriculum. It is not a single capability but a concoction of capabilities working in tandem. Abilities like precision, accuracy, managing impulsivity, lateral thinking, persistence, applying past knowledge to new situations, and lastly, retention and recollection etc., need to be worked in unison to solve problems.

Here is a life experience from the past. Class IX witnessed the introduction of quadratic equations, and my elder one used to get most of the answers wrong. There was nothing besides addition, subtraction, and multiplication required to solve these problems, so the volume of errors was puzzling. JoveNotes, via its analytics, hinted that the issue lay in lack of perseverance, fueled by the inability to manage impulsivity. So instead of prodding him to do more quadratic problems, I put a 350-degree soldering iron in his hands and encourage him to play with electronics. Routing connections on a Veroboard taught him perseverance, and soldering helped him manage impulsivity. The enthusiasm for playing with electronics overshadowed the pain of blisters. Math has not been haunting him ever since.

There was another serious problem. Problem-solving requires constant practice; it's not a memory game. Because of our sequential approach to studies, there is an inevitable rush near to the exams. Those topics taught earlier in the academic season usually don't get enough revision.

The collage consists of five screenshots from the JoveNotes application, each marked with a purple circle containing a number:

- 1:** A screenshot of the 'Exercises' dashboard. It shows a list of exercises categorized by subject (Mathematics) and topic (e.g., Large Numbers, Addition Subtraction, Multiplication Division). Each exercise has a progress bar and a 'Start Exercise' button.
- 2:** A screenshot of the 'Start Exercise (6)' screen. It displays a table of exercises with columns for Subject, Exercise, Meter, and Start. A 'Configure exercise' button is visible at the bottom.
- 3:** A screenshot of the 'Root cause attribution' screen. It shows a grid of icons representing different cognitive skills or attributes, such as 'Managing impulsivity', 'Flexibility', and 'Responding with wonderment and awe'. An 'Apply' button is at the bottom right.
- 4:** A screenshot of the 'Exercise (6 questions) - Solve Phase' screen. It displays a math problem: 'Add and subtract the following. (Do these sums in your notebook.)' followed by the calculation $1,89,774 + 4,55,000 = 3,96,719$. Below the problem is a subtraction problem: $879658 - 64548$. The screen includes a timer and buttons for 'Answer' and 'Next'.
- 5:** A screenshot of the 'Performance Summary' screen. It shows a table with columns for Question ID, Level (easy, medium, hard), Level (good, better, great), Difficulty, Time spent, Rating, and Score. The table lists questions Q1 through Q6. A 'Total score' of 180 is shown at the bottom. A 'Return to exercise dashboard' button is at the bottom right.

JoveNotes approaches problem-solving in a novel way. It allows choosing a set of topics (screen 1) to assembly a practice session (screen 2). Students can construct practice sessions by selecting different questions from multiple chapters. The platform then conducts the exercise by guiding through various stages (screen 4), like observing, thinking, attempting, re-attempting, and revising. Once the practice is over, JoveNotes helps establish root causes for errors (screen 3 and 5) which can be further analyzed and fed into the mentoring process.

An awesome journey

For me, this journey has been a rewarding experience both as a mentor and as a software engineer. With eight years of active development and more than 2000 GitHub commits, this will also be the longest-running software project of my career till now.

While JoveNotes played a centerstage role in our K10 studies, it could not lend its help beyond the 10th. So, I had to build another platform specifically for JEE preparation, named SConsole. SConsole, shown below, has been instrumental in helping my son prepare for his JEE exams without the need for any external coaching. SConsole is a story for another day, though.

